01. Environment Setup

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Introduction

Before we write a single line of Python, we need to build our "Workbench."

In the world of Big Data, **version control** is critical. Apache Spark (the tool you will learn later) is very picky about which version of Python and Java it uses. If you just install the latest Python from the internet, Spark might fail.

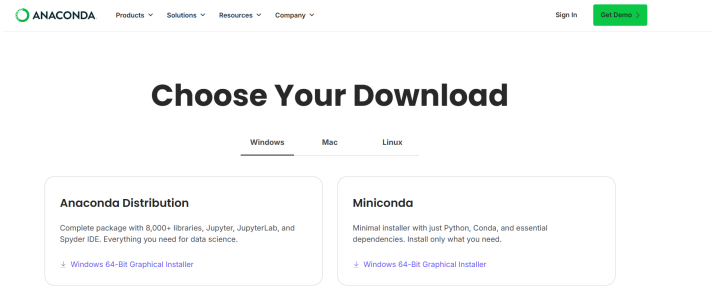
Our Goal:

1. Use **Miniconda** to manage our software (it allows us to switch Python versions easily).
2. Create a specific environment named `bigdata` with **Python 3.10** (perfect for Spark 3.x).
3. Connect **PyCharm** (our code editor) to this environment.

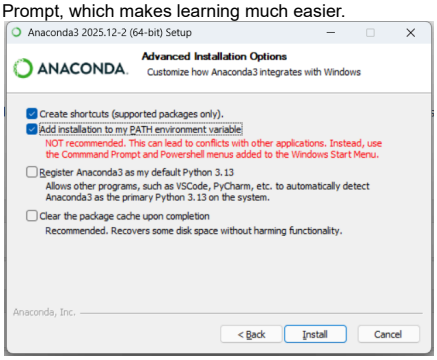
Part 1: Installing the Manager (Miniconda)

Think of Miniconda as a toolbox. Right now, it's empty, but it gives us the ability to create separate "compartments" (environments) for different projects.

1. **Download Miniconda**
   - Visit the Miniconda Download Page.
   - Download the **Python 3.x (64-bit)** installer for your Operating System (Windows, macOS, or Linux).



2. **Run the Installer**
   - **Windows Users (Crucial Step):** During installation, you will see a checkbox that says *"Add Miniconda3 to my PATH environment variable."*
     - The text is red and says "Not Recommended," but for this course, **please CHECK this box.** It will allow you to use Conda commands directly in your Command Prompt, which makes learning much easier.



   - **Mac/Linux Users:** Just follow the standard prompts (Agree -> Install).
3. **Verify Installation**
   - Open your terminal (Windows: Search for `cmd`; Mac: Search for `Terminal`).
   - Type:

```
conda -V
```

   - *Success:* If you see something like `conda 23.x.x`, you are ready.

```
conda 25.11.1
```

Part 2: Building the "Big Data" Environment

Now we will create a quarantined environment. We will not use the "base" Python. We will build a custom one.

1. **Open your Terminal/Command Prompt.**
2. **Create the Environment:** Type the following command and press Enter. This tells Conda to download Python 3.10.

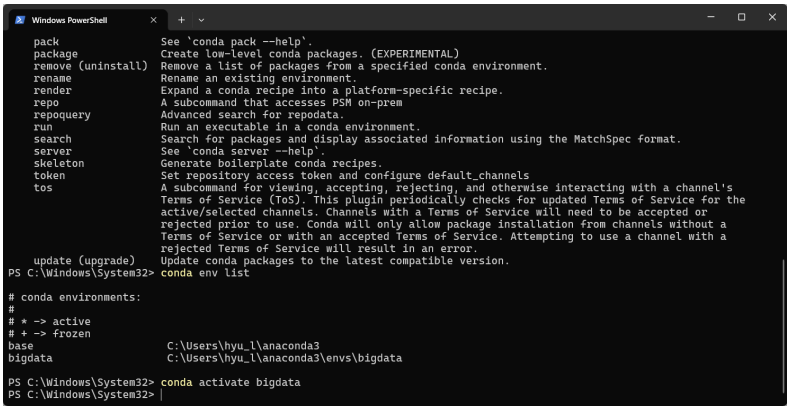
```
conda create -n bigdata python=3.10 -y
```

(Note: `-n bigdata` names the environment. `python=3.10` ensures compatibility with Spark 3.x)

3. **Activate the Environment:** We need to step *inside* this new environment.

```
conda activate bigdata
```

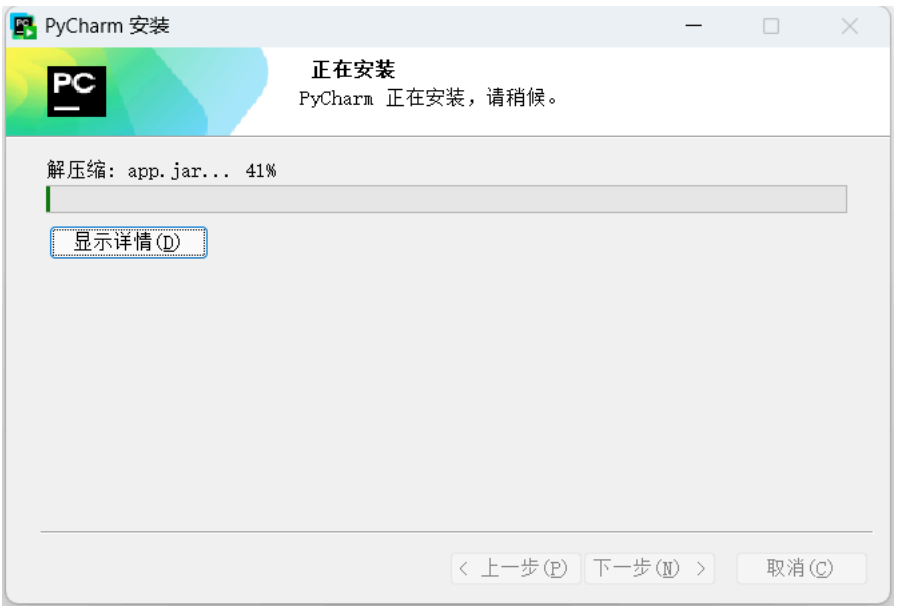
(Notice: Your terminal prompt should change from `(base)` to `(bigdata)`).



Part 3: Installing the Editor (PyCharm)

You need a good IDE (Integrated Development Environment). PyCharm is the industry standard.

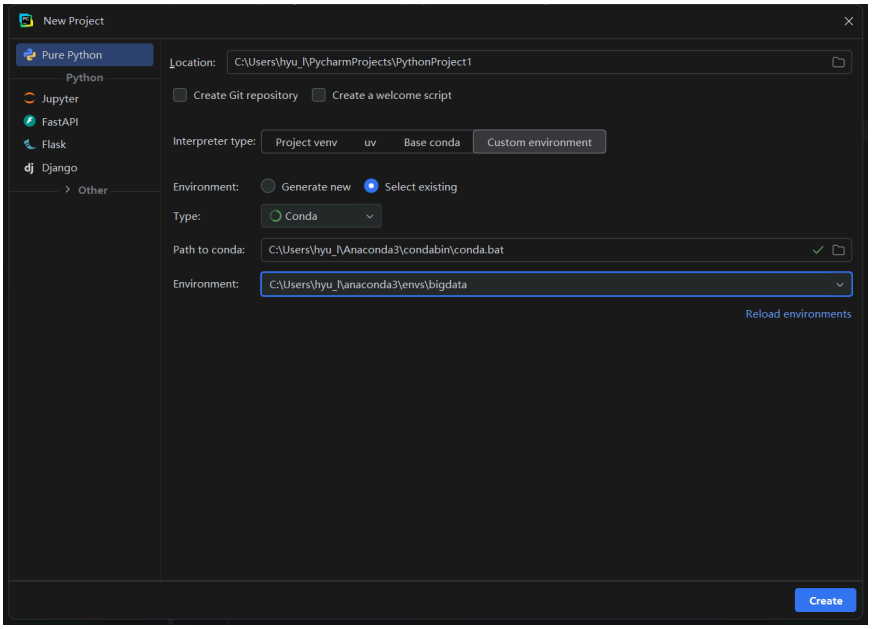
1. **Download PyCharm:**
   - Go to jetbrains.com/pycharm/download.
   - **Scroll down** to download based on your system.
   - Download and install it..



Part 4: Connecting the Wires

This is the most important step. We need to tell PyCharm: *"Don't use the computer's default Python. Use the 'bigdata' environment we just made."*

1. **Create a New Project:**
   - Open PyCharm.
   - Click **New Project**.
   - **Location:** Name your folder (e.g., `python-for-bigdata`).
2. **Configure the Interpreter (The Connection):**
   - Look for the **Python Interpreter** dropdown menu.
   - Select **Custom Environment** (or "Previously Configured Interpreter").
   - Click **Add Interpreter** (or the generic "..." button).
   - Select **Conda Environment**.
3. **Select Existing Environment:**
   - **Do not** select "New environment." Select **Load Existing Environment** (or "Use Existing").
   - In the "Interpreter" dropdown, look for your `bigdata` environment.
   - *Troubleshooting:* If you don't see it, you may need to find the file manually:
     - **Windows:** `C:\Users\YOUR_NAME\miniconda3\envs\bigdata\python.exe`
     - **Mac:** `/Users/YOUR_NAME/opt/miniconda3/envs/bigdata/bin/python`
4. Click **Create**.



Part 5: The "Hello World" Verification

Let's prove everything is working.

1. In PyCharm, right-click your project folder (left sidebar) → **New** → **Python File**.
2. Name it: `setup_check.py`.
3. Paste the following code:

```
import sys

def check_environment():
    print("--- Checking Python Version ---")
    print(f"Current Version: {sys.version}")

    if "3.10" in sys.version:
        print("✅ Python version is CORRECT for Spark.")
    else:
        print("⚠️ Warning: Version might be incompatible. Spark is better running under 3.10 ")

if __name__ == "__main__":
    check_environment()
```

4. Right-click anywhere in the code window and select **Run 'setup_check'**.

What you should see in the console:

```
--- Checking Python Version ---
Current Version: 3.10.19 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 16:41:31) [MSC v.1929 64 bit (AMD64)]
✅ Python version is CORRECT for Spark.
```

Conclusion

Congratulations! You have successfully set up a professional-grade Big Data development environment.

- You have **Python 3.10** (Logic layer).
- You have **PyCharm** (Development layer).

No labels